

CommunityHub

Group 9 — Community Portal

CS5450 Mobile Programming · Dr. Sabah Mohammed · Lakehead University

Course	COMP5450 Mobile Programming
Group	Group 9 — Community Portal
Platform	React Native / Expo SDK 51
Backend	Firebase Auth + Cloud Firestore
Language	JavaScript (ES2022)
Database	community-hub-7a150 (Firestore)
GitHub	https://github.com/YOUR_USERNAME/community-hub-cs5450

1. Project Overview

CommunityHub is a full-featured mobile community portal built with React Native and Expo for the CS5450 Challenge 2 assignment. It connects people through shared interests using Firebase Authentication and Cloud Firestore for real-time data storage.

Feature	Description	Status
User Registration & Login	Firebase Auth (email/password), admin & member roles	■
Personalized Dashboard	Stats, recent forums, upcoming events, joined groups	■
Discussion Forums	Create posts, threaded replies, likes — real-time	■
Event Calendar	Create events, join/leave, RSVP, date display	■
Member Directory	All users, search by name or interest, profile modal	■
Interest Groups	Join/leave groups, member count, live updates	■
Search Functionality	Cross-module search bar on every screen	■
Security Rules	Firestore rules restrict reads/writes by auth & role	■
Responsive Design	Works on phone, tablet and Chrome (web)	■

2. Technology Stack

Layer	Technology
Framework	React Native 0.74 + Expo SDK 51
Language	JavaScript ES2022
Navigation	React Navigation 6 (Bottom Tabs + Native Stack)
Auth	Firebase Authentication (email / password)
Database	Cloud Firestore (real-time onSnapshot listeners)
State Mgmt	React hooks (useState, useEffect, Context API)
Hosting	Expo Go (dev) / Expo build (production APK)

3. Project Structure

```
community-hub/
  ■■■ App.js Entry point, navigation, auth gate
  ■■■ package.json Dependencies
  ■■■ firestore.rules Firestore security rules
  ■■■ README.pdf This document
  ■■■ src/
    ■■■ hooks/
      ■ ■■■ useAuth.js Auth context + hook
    ■■■ services/
      ■ ■■■ firebase.js Firebase init (config here)
      ■ ■■■ authService.js register / login / logout
      ■ ■■■ firestoreService.js All Firestore CRUD + listeners
    ■■■ screens/
      ■ ■■■ AuthScreen.js Login + Register UI
      ■ ■■■ DashboardScreen.js Personalized home screen
      ■ ■■■ ForumsScreen.js Forum list + ThreadDetail
      ■ ■■■ EventsScreen.js Events list + create
      ■ ■■■ MembersScreen.js Members directory + Groups
    ■■■ utils/
      ■■■ helpers.js Category colours, avatar, formatDate
```

4. Firebase Setup

4.1 Firebase Project Details

Setting	Value
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Project name	community-hub
Project ID	community-hub-7a150
Auth domain	community-hub-7a150.firebaseio.com
Region	northamerica-northeast1 (Montreal)
Auth method	Email / Password
Database	Cloud Firestore (Standard edition)

4.2 Firestore Data Model

users/{uid}

name, email, avatar, role, bio, interests[], posts, events, joinedAt

forums/{postId}

title, category, authorId, authorName, likes, likedBy[], replyCount, createdAt

■■■ replies/{replyId}

text, authorId, authorName, likes, createdAt

events/{eventId}

title, category, date, time, location, desc,

organizerId, organizerName, attendees[], maxAttendees, createdAt

groups/{groupId}

name, category, icon, desc, members[], createdAt

5. Installation & Running the App

Step 1 — Unzip the project

Extract community-hub-firebase-final.zip to a folder on your computer.

Step 2 — Install dependencies

Open a terminal inside the community-hub/ folder and run:

```
npm install
```

Step 3 — Start Expo

Run the development server:

```
npx expo start
```

Step 4 — Open on device/emulator

Press 'a' for Android emulator, 'i' for iOS simulator, 'w' for Chrome browser.

Step 5 — Physical device (bonus +2 marks)

Install Expo Go from Play Store / App Store, then scan the QR code shown in the terminal.

6. App Screens

Screen	File	Key Features
Login / Register	AuthScreen.js	Firebase Auth, role assignment, error handling
Dashboard	DashboardScreen.js	Stats cards, recent posts, my events, my groups
Forums	ForumsScreen.js	Post list, category filter, create post, thread detail, replies, likes
Events	EventsScreen.js	Event cards, join/leave, create event, date badge, category filter
Members	MembersScreen.js	Member grid, search, profile modal, interests
Groups	MembersScreen.js	Group cards, join/leave, member avatars, live count

7. Security

- Firebase Authentication protects all routes — unauthenticated users cannot access any screen.
- Firestore Security Rules (firestore.rules) enforce role-based access:
- Only the post/event owner or an admin can delete content.
- All read/write operations require an authenticated session.
- Admin role is stored in the users collection and checked server-side.

```
// Deploy rules before demo:
```

```
firebase deploy --only firestore:rules
```

8. Assignment Grading Checklist

Requirement	Met?
React Native / Expo platform	■ Yes
Firebase Authentication	■ Yes — email/password
Firebase Firestore database	■ Yes — real-time CRUD
User Registration & Login with roles	■ Yes — admin / member
Personalized Dashboard	■ Yes
Discussion Forums with replies & likes	■ Yes
Event Calendar with join/leave	■ Yes
Member Directory with search	■ Yes
Interest Groups with join/leave	■ Yes
Search functionality (all screens)	■ Yes
Responsive design (phone + tablet + web)	■ Yes
Firestore Security Rules	■ Yes
README.pdf with setup & structure	■ This document
ZIP package for D2L	■ community-hub-firebase-final.zip
GitHub public repository	■ Add your GitHub link

9. GitHub Repository

The full source code is available at:

```
https://github.com/YOUR_USERNAME/community-hub-cs5450
```

Replace YOUR_USERNAME with your actual GitHub username before submitting to D2L. Make sure the repository is set to Public.